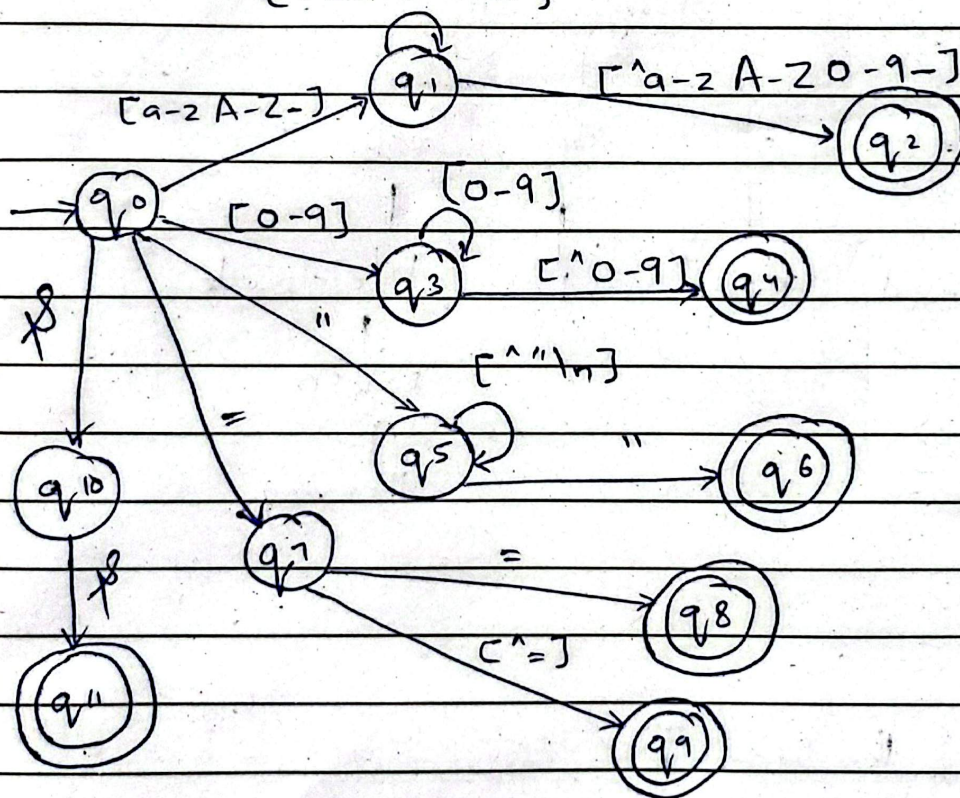


Lexical Analyzer Design

Regular Expression = $[a-zA-Z_][a-zA-Z0-9_]* \mid [0-9]^+ \mid$
 $"[^"\\n]^*" \mid == \mid [=] \mid \text{space}$
 $[a-zA-Z0-9_]$

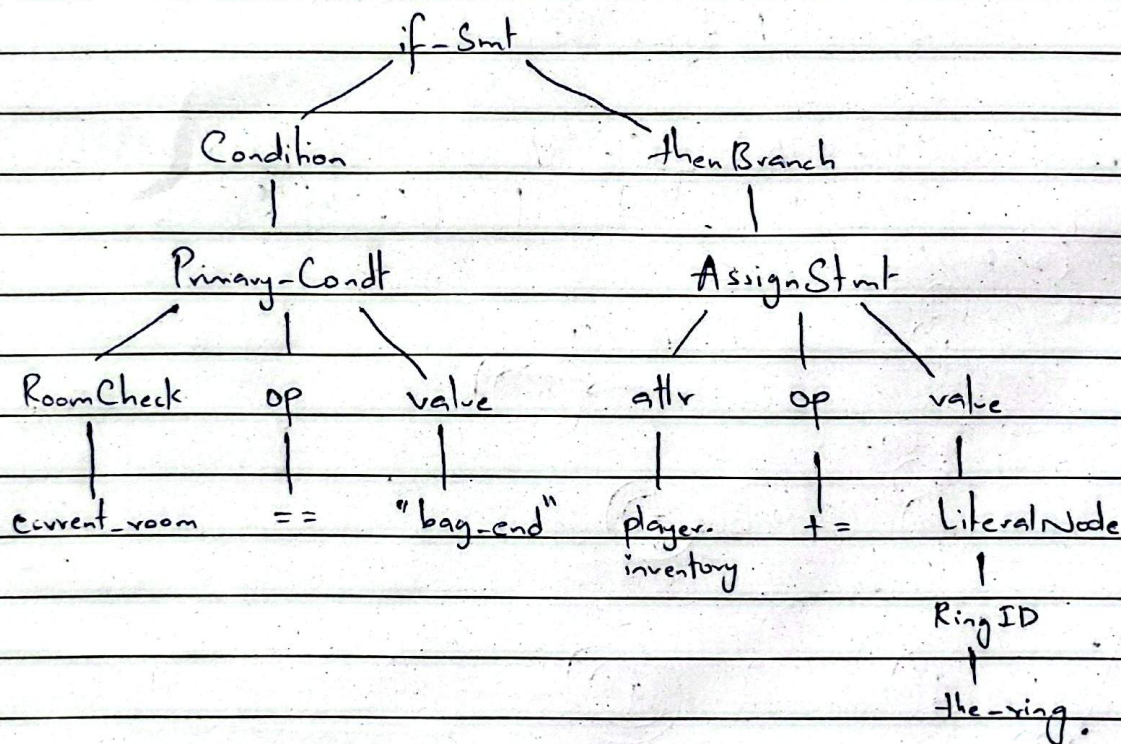


Date: _____

Parse Tree

Tree #1

if current_room == "bag-end" { player.inventory += the-ring }



Production Rules

if-stmt $\rightarrow$ if Condition { then Branch }

Condition $\rightarrow$ Primary-Cond

Primary-Cond $\rightarrow$ RoomCheck == STRING

thenBranch $\rightarrow$ Assign Stmt

Assign Stmt $\rightarrow$ attr op value

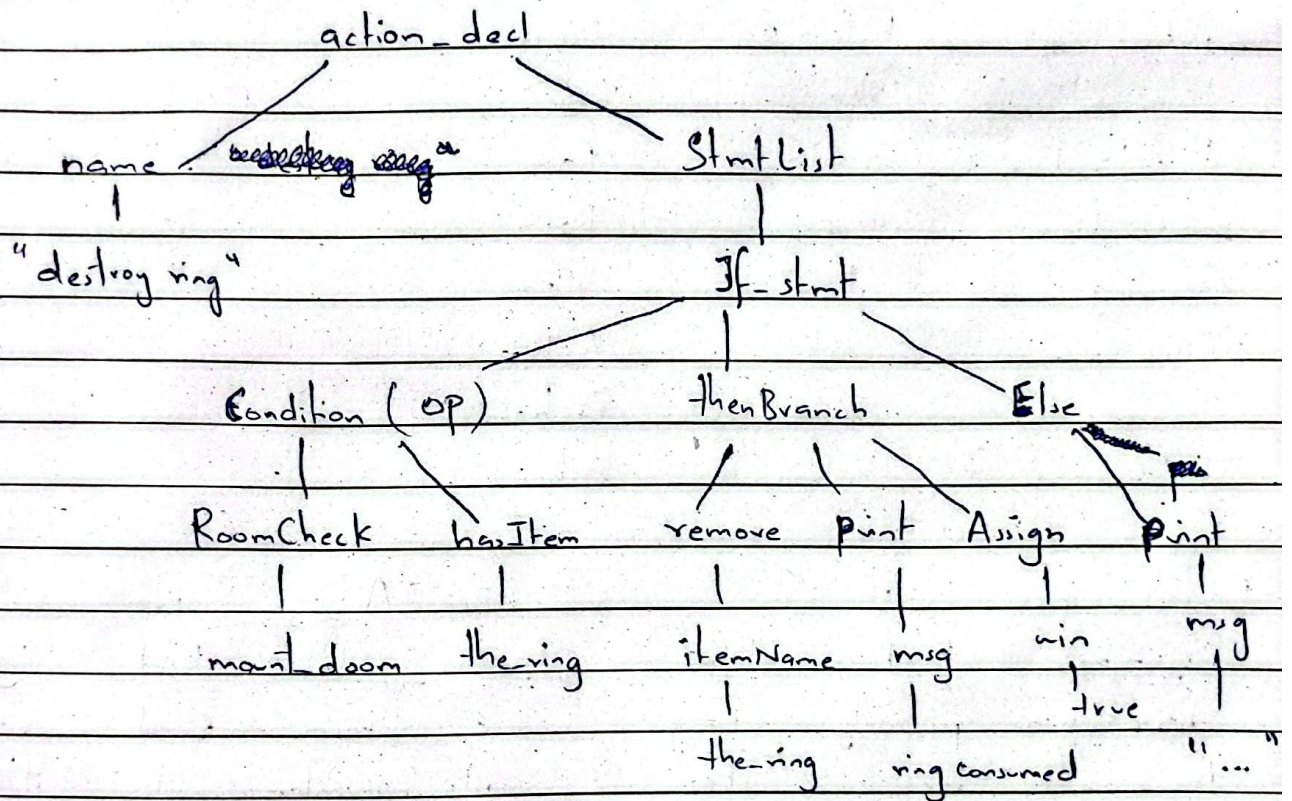
attr $\rightarrow$ ID.ID

op $\rightarrow$ +=

value $\rightarrow$ LiteralNode

LiteralNode $\rightarrow$ ID

Tree #2 action "destroy ring" { if current_room == "mount-doom"
 { player.has-item(the-ring) { remove the-ring
 print "...". player.win == true } else { print "..."} }



Production Rules

action_decl $\rightarrow$ "action" STRING "{ " statement " } "

If-stmt $\rightarrow$ "if" condition "{ " statement " } " thenBranch else

Condition $\rightarrow$ RoomCheck has-Item

thenBranch $\rightarrow$ remove print Assign

Else $\rightarrow$ print

print $\rightarrow$ msg

msg $\rightarrow$ "..." | STRING

• Semantic Analyzer

Global Scope

<u>Symbol</u>	<u>Kind</u>	<u>Type</u>
current-room	builtin var	room-ref
player	builtin obj	Player
bag-end	room	Room
buckland	room	Room
mount-doom	room	Room
the-ring	item	Item
"take ring"	action	Action
"examine ring"	action	Action
"destroy ring"	action	Action

Room Scopes

<u>Symbol</u>	<u>Kind</u>	<u>Type</u>
description	property	string
the-ring	item	Item
exit.east	exit	room-ref
exit.west	exit	room-ref
health	attribute	int
hostile	attribute	bool
sauron	npc	NPC
power	attribute	int
type	attribute	string

Action Scopes

<u>Symbol</u>	<u>Kind</u>	<u>Type</u>
current-room	read	room-ref
player.inventory	write	item[]
print	builtin call	string
player.has-item()	builtin item	bool
player.win	write	bool
remove	builtin call	item(void)